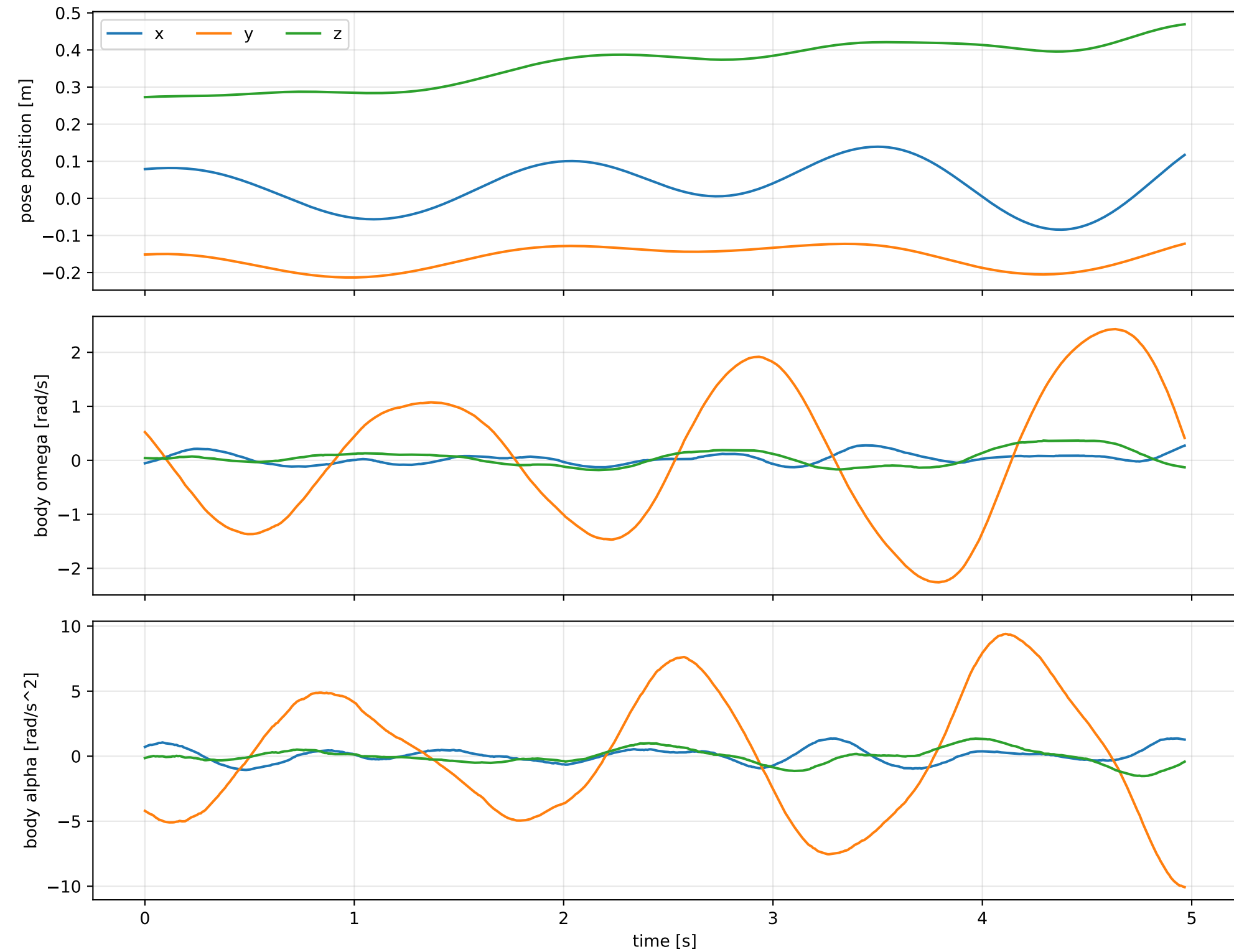
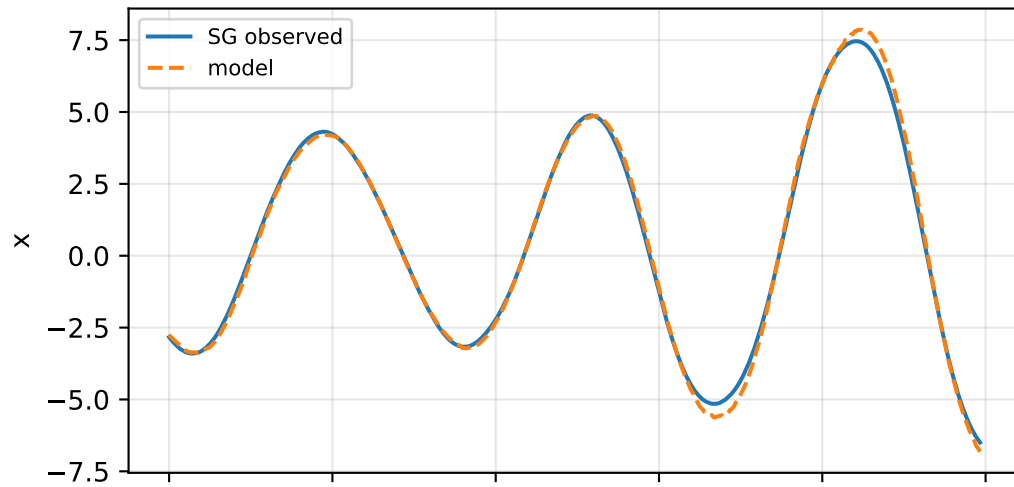


kkt_scale_positive: SG trajectory and derived motion

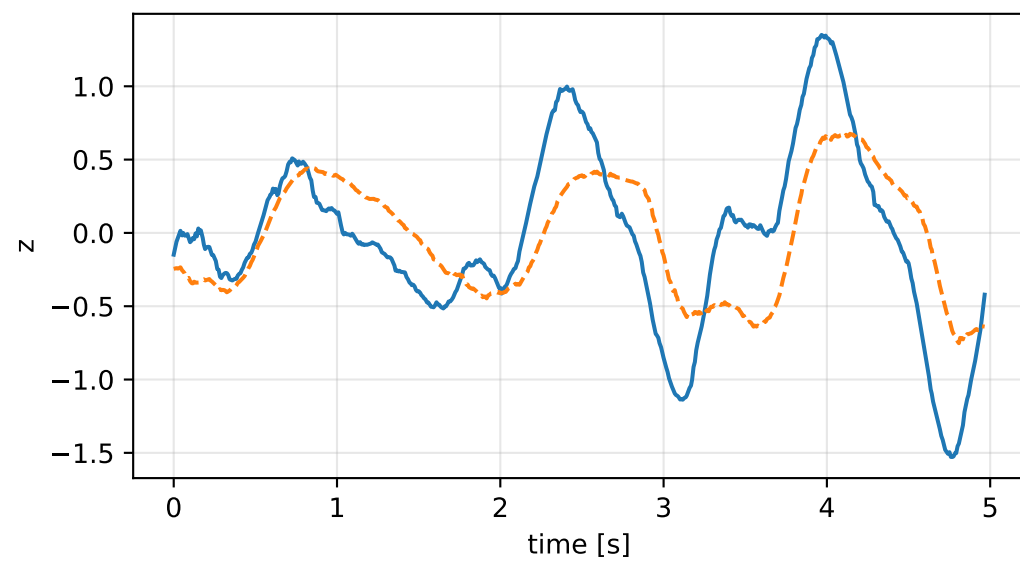
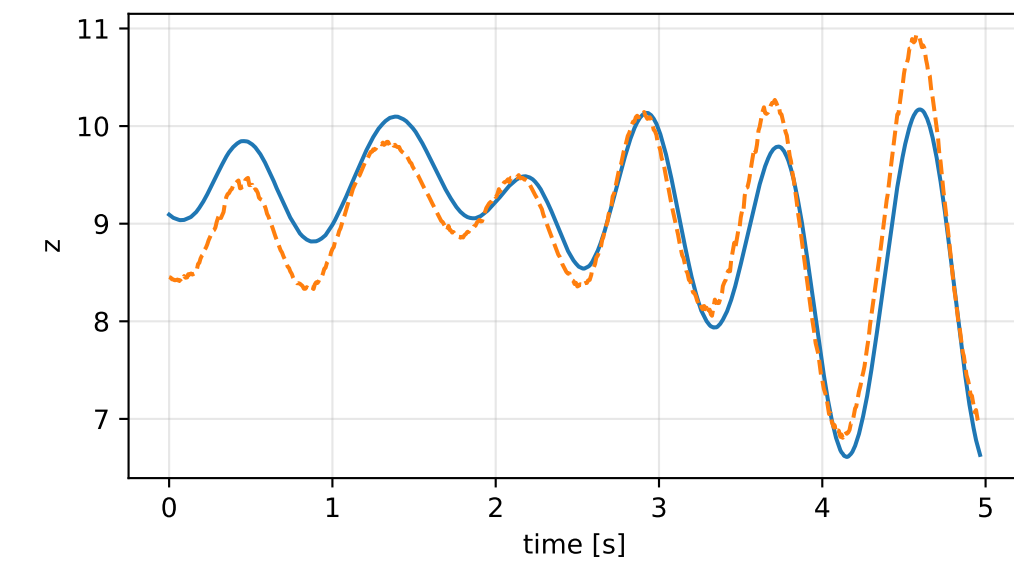
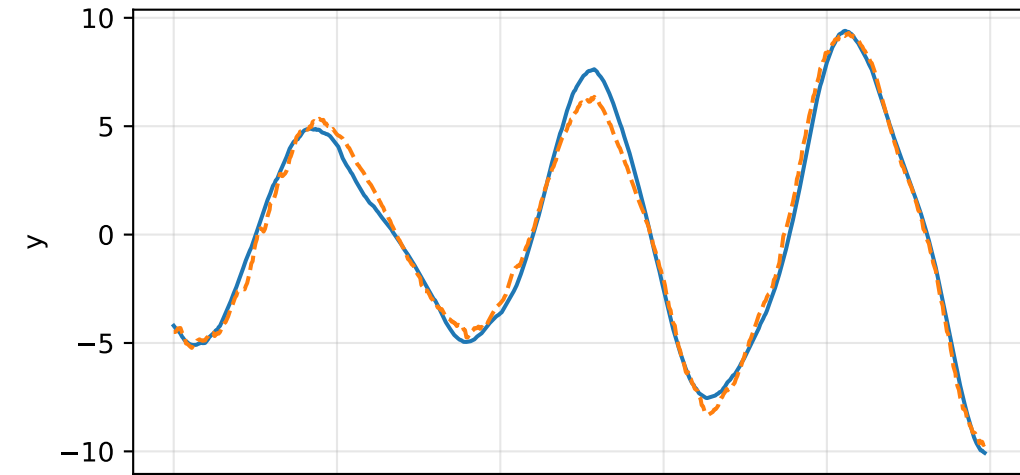
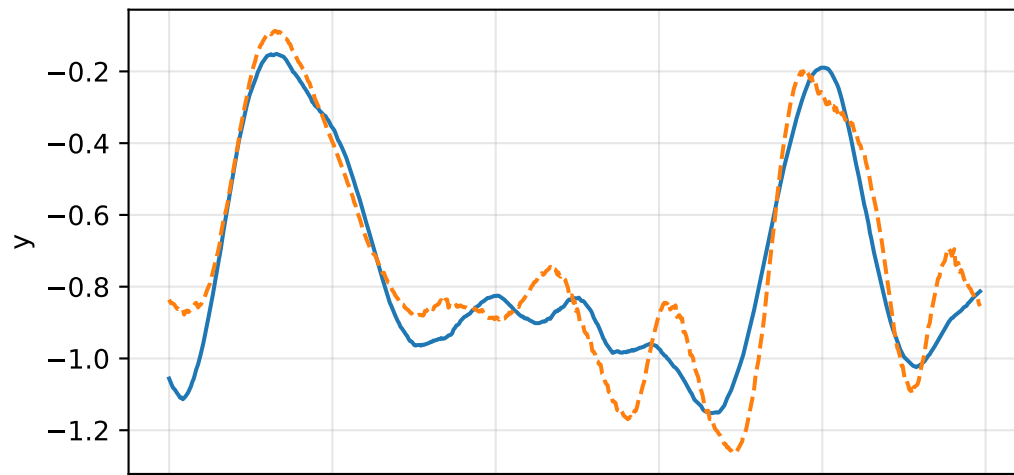
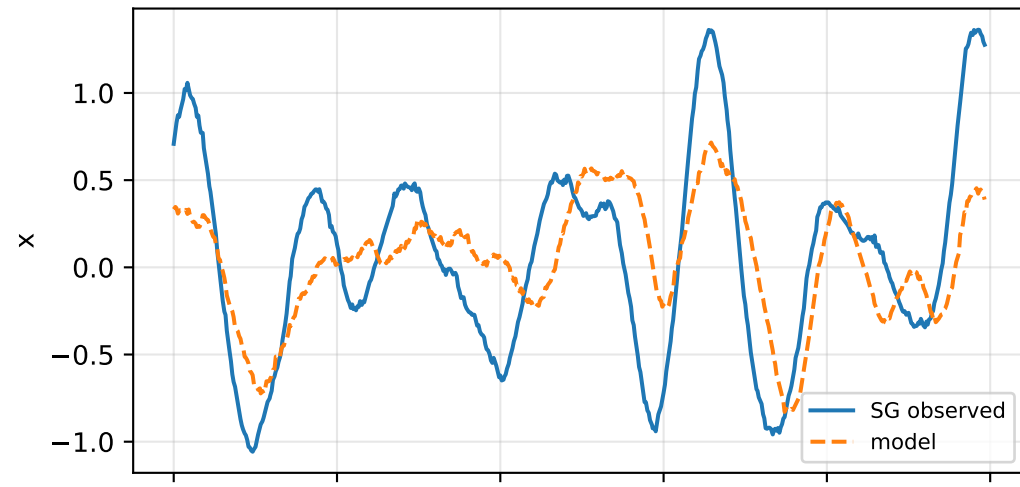


kkt_scale_positive: acceleration residual objective

specific acceleration [m/s^2]

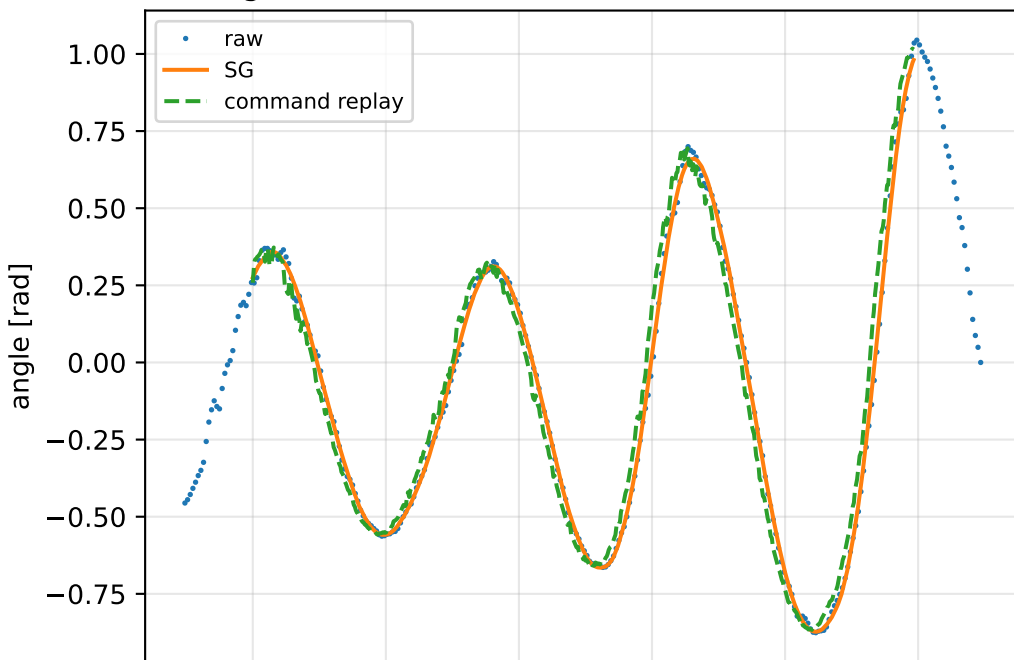


angular acceleration [rad/s^2]

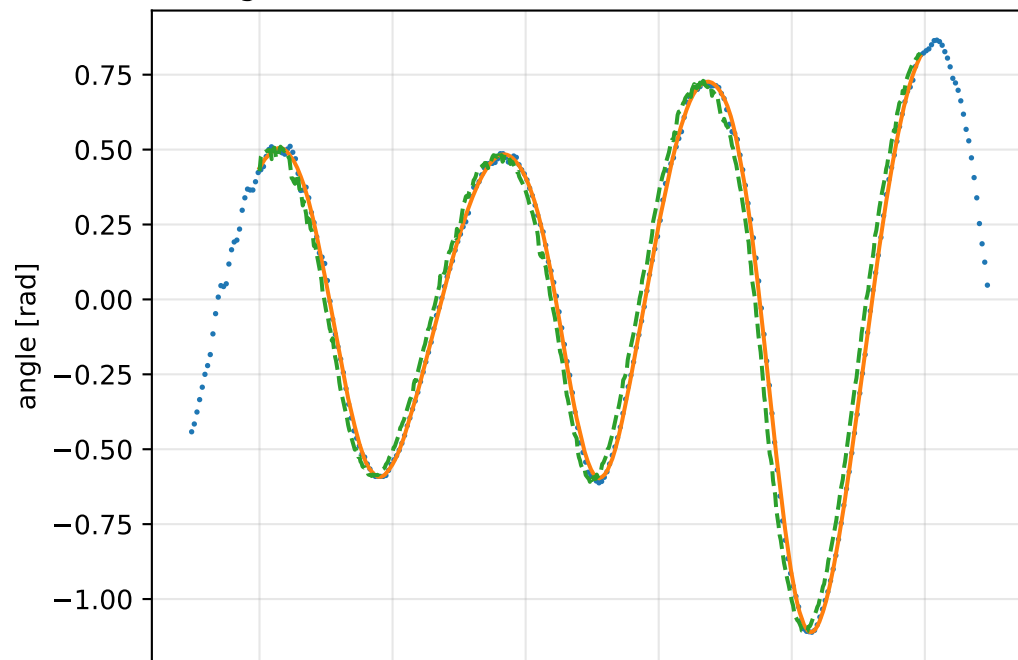


kkt_scale_positive: gimbal measurement audit (objective=measured_sg)

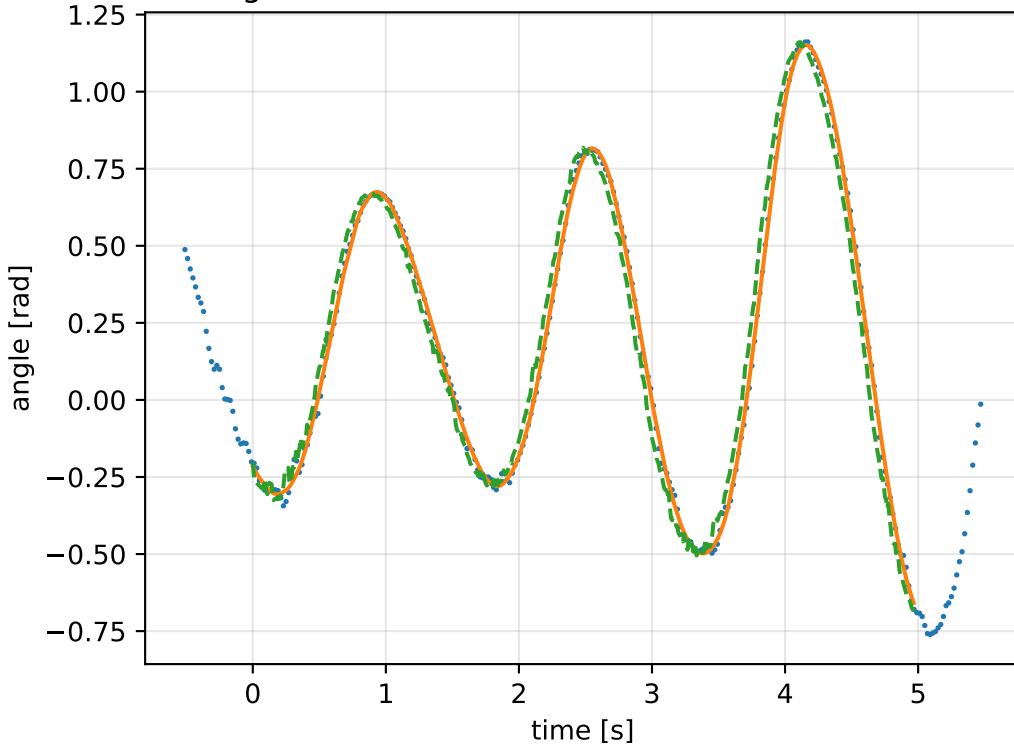
gimbal 1: RMSE=0.0759, max=0.2095 rad



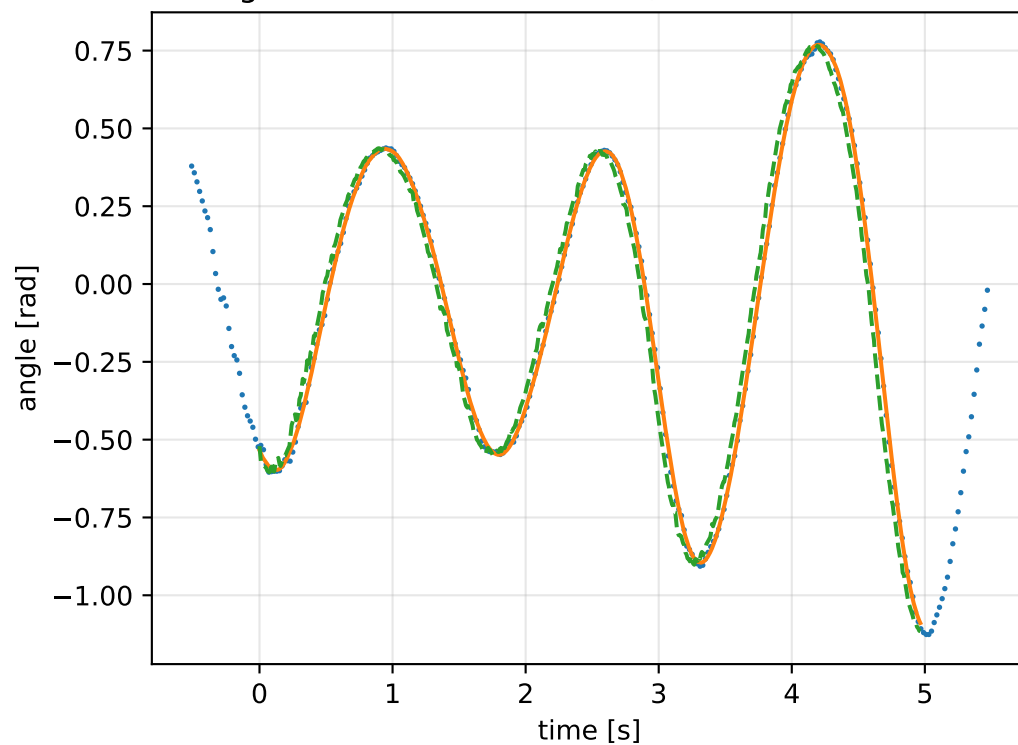
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

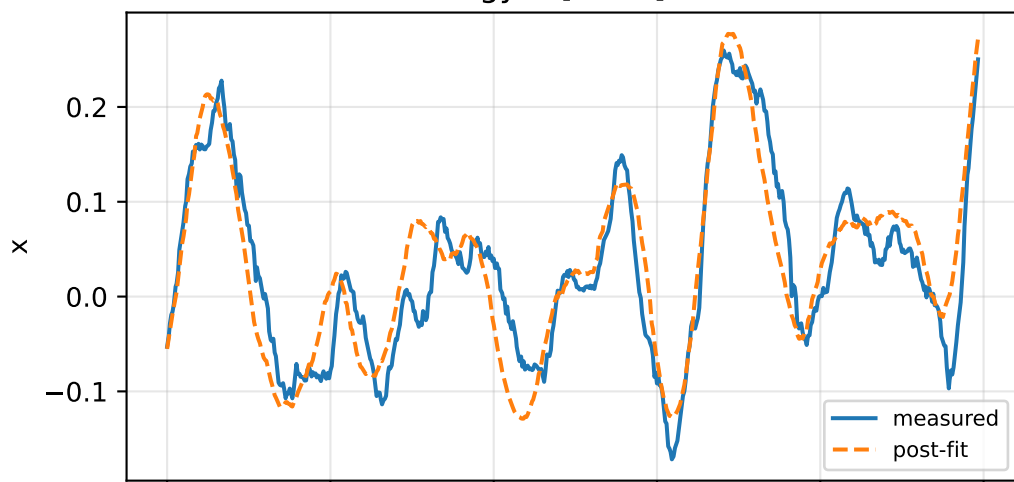


gimbal 4: RMSE=0.07131, max=0.1737 rad

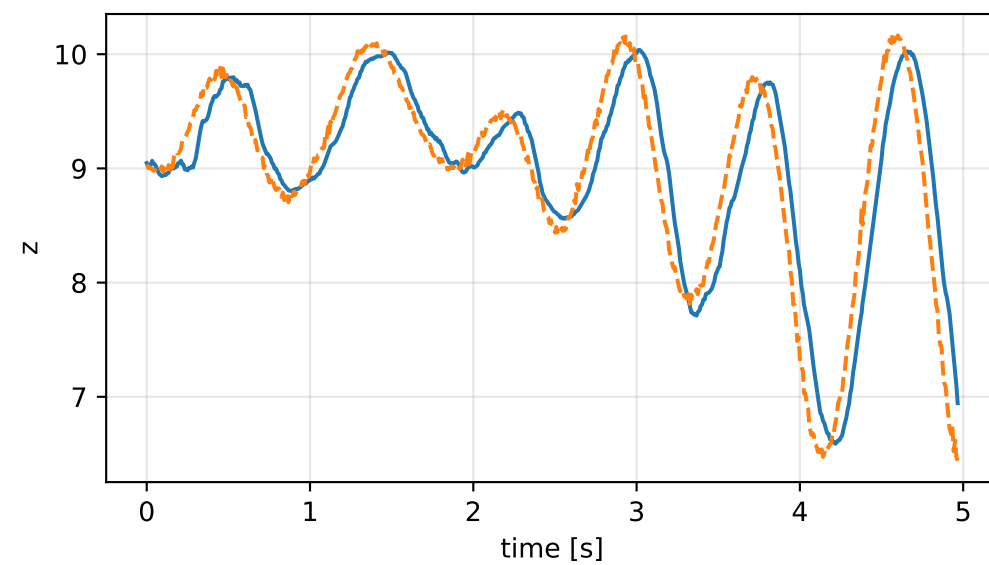
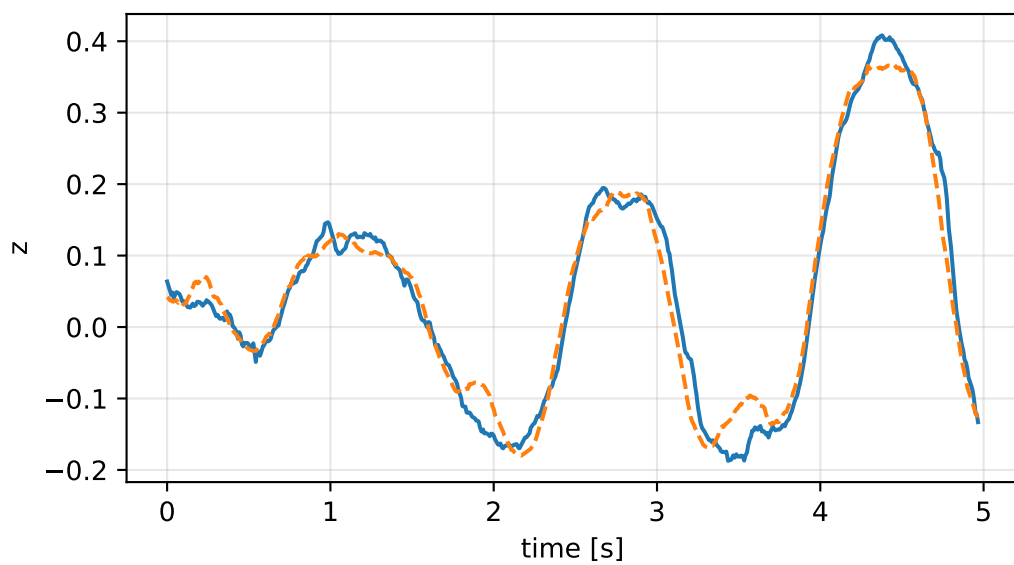
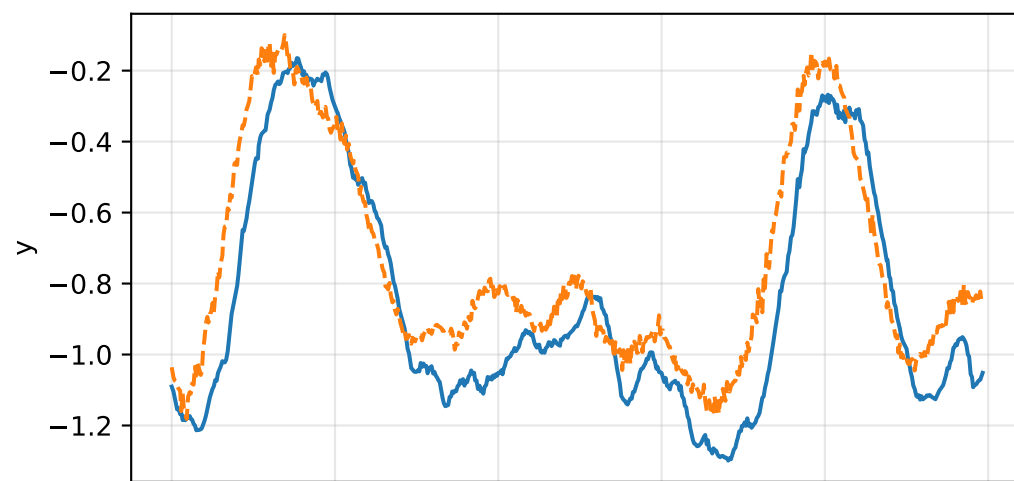
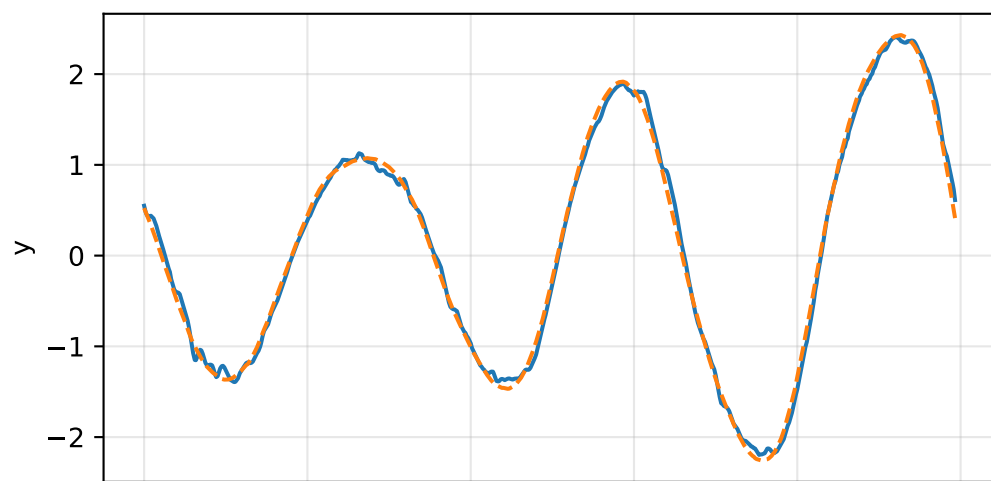
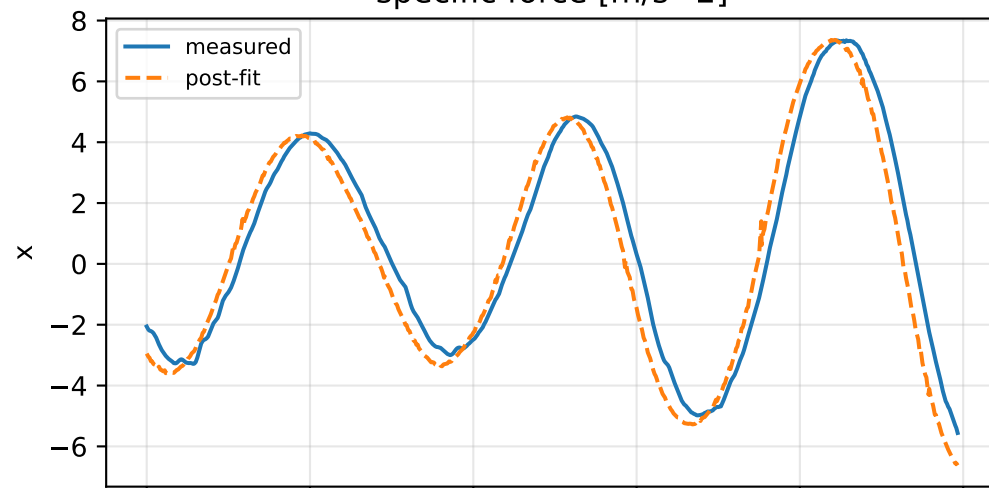


kkt_scale_positive: IMU comparison (diagnostic only)

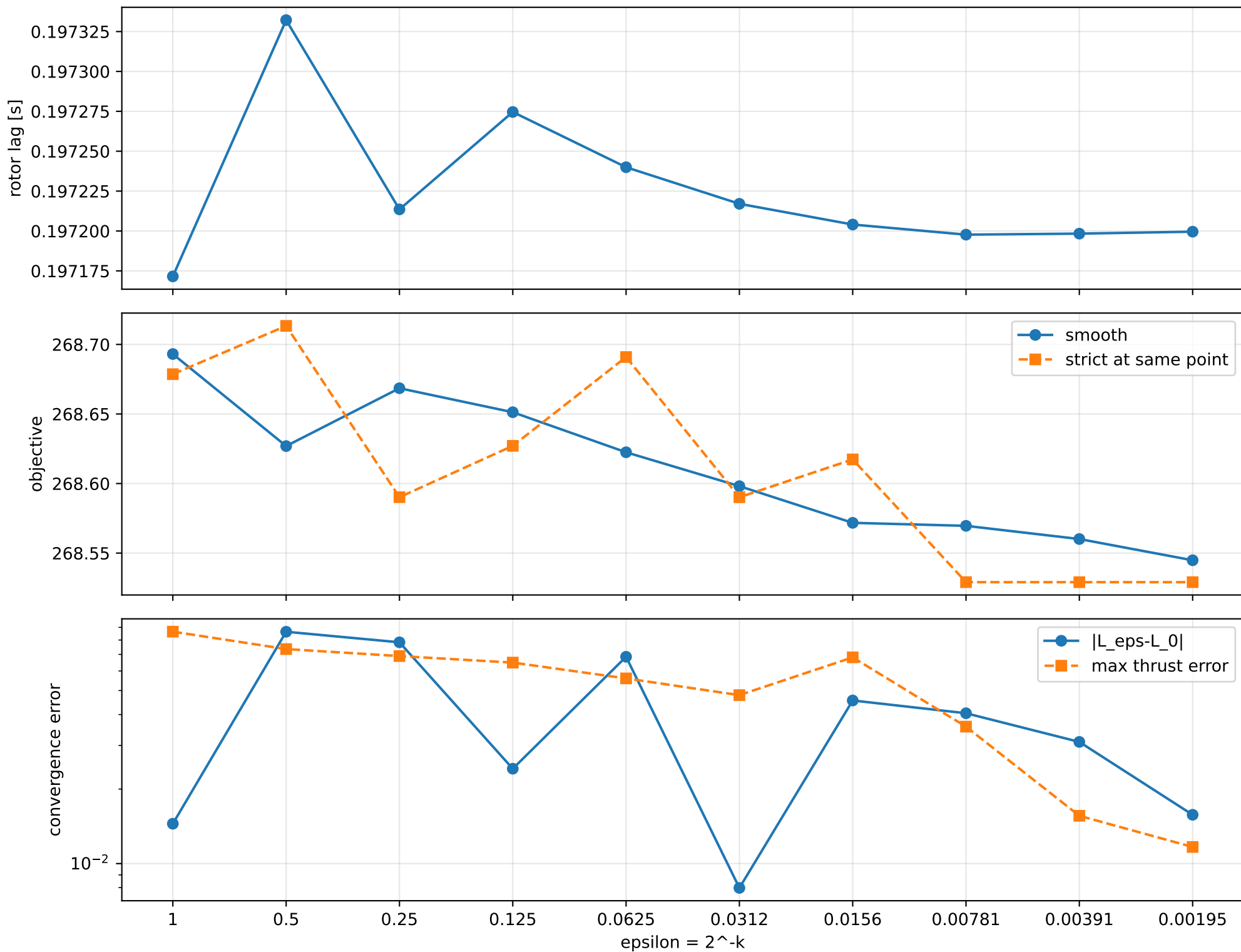
gyro [rad/s]



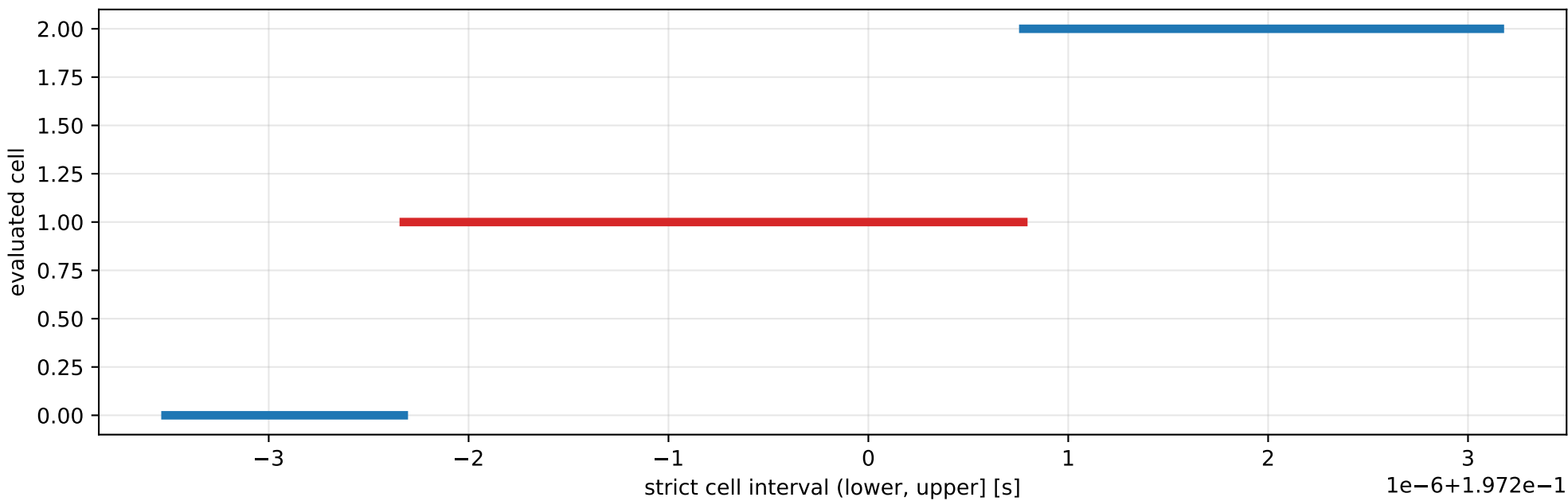
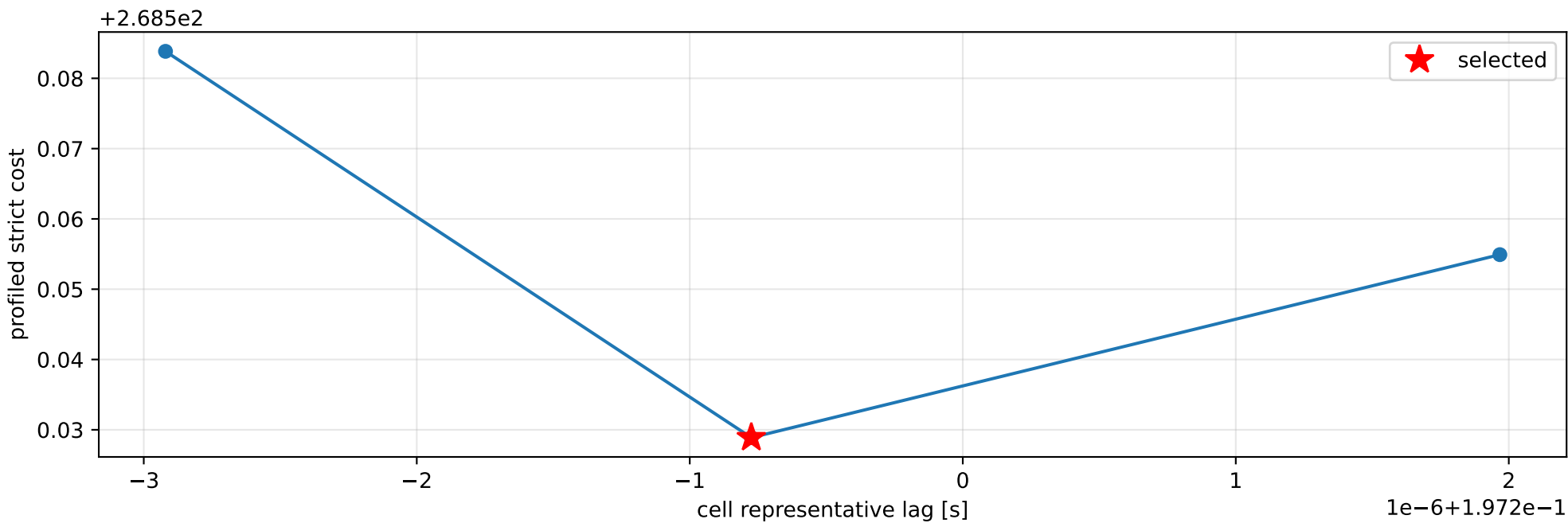
specific force [m/s^2]



kkt_scale_positive: smooth-to-strict continuation



kkt_scale_positive: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199564 s; data boundary=False

kkt_scale_positive: parameter estimate

Case: kkt_scale_positive

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.47573919

inertia [kg m²]:

[[3.53067464e-01 -4.27100120e-04 -5.73203733e-02]

[-4.27100120e-04 2.68867277e-01 3.30980041e-05]

[-5.73203733e-02 3.30980041e-05 5.96255973e-01]]

CoG body [m]: [0.01762808 -0.00411735 -0.04437562]

force effectiveness: [1.16405654 1.49895804 1.30895698 1.08525949]

scale-free J/m [m²]:

[[1.01580540e-01 -1.22880371e-04 -1.64915634e-02]

[-1.22880371e-04 7.73554236e-02 9.52257991e-06]

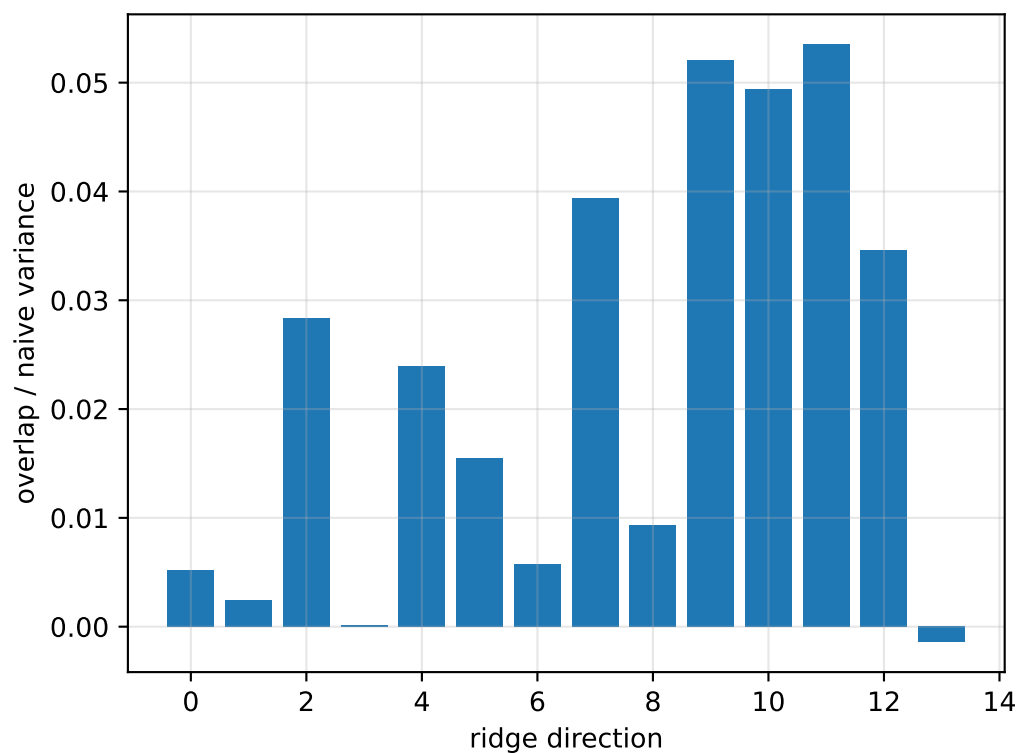
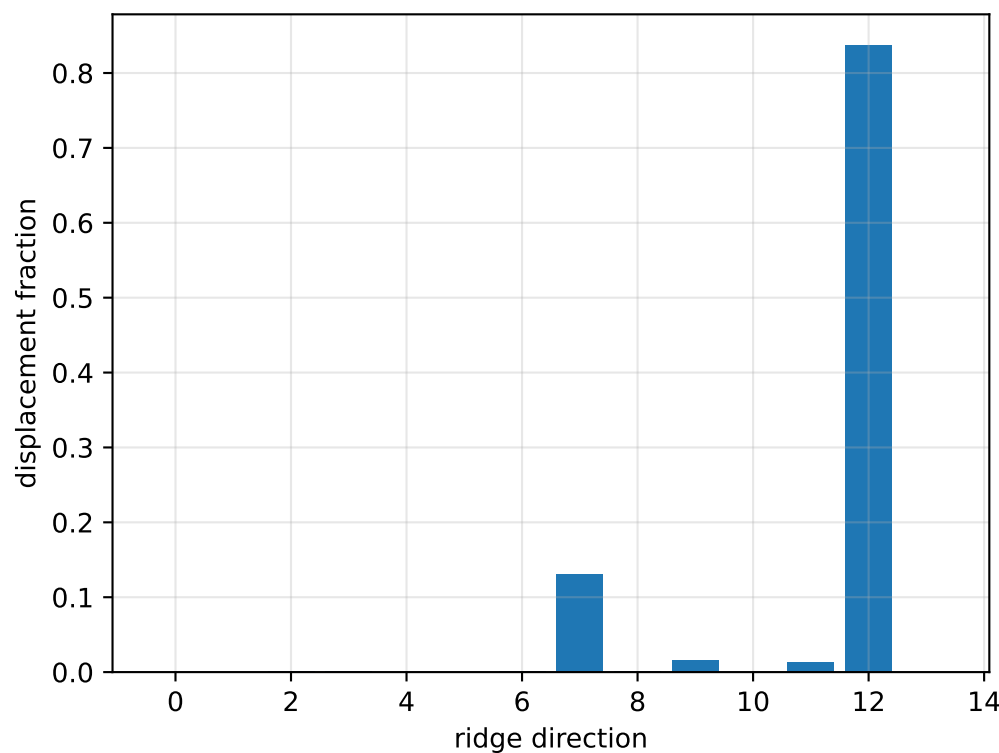
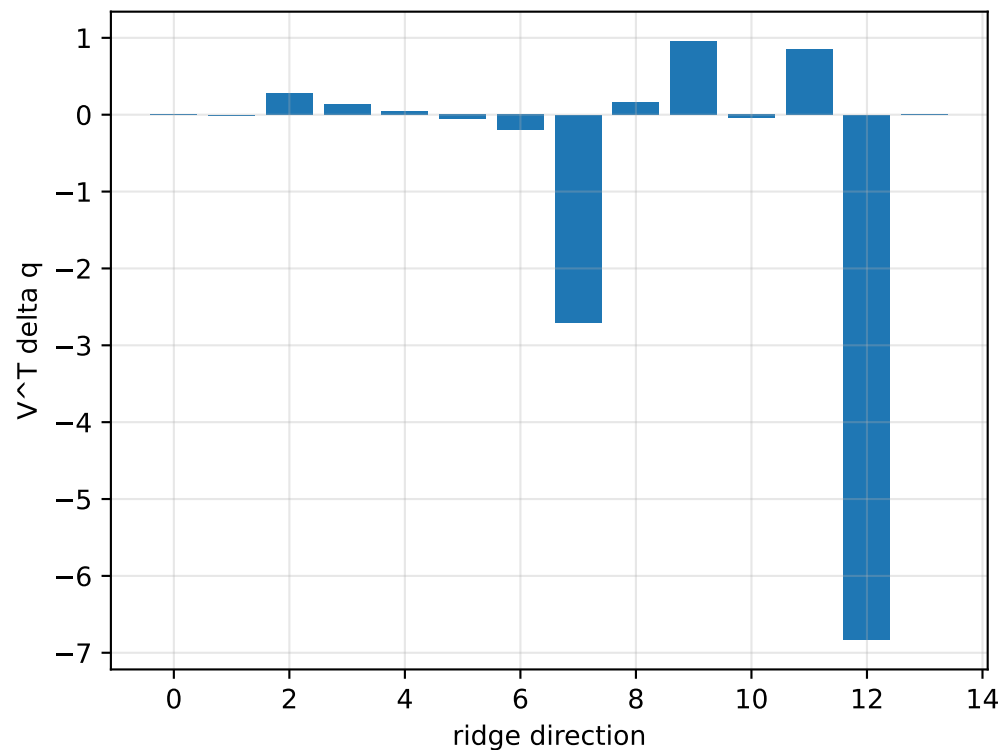
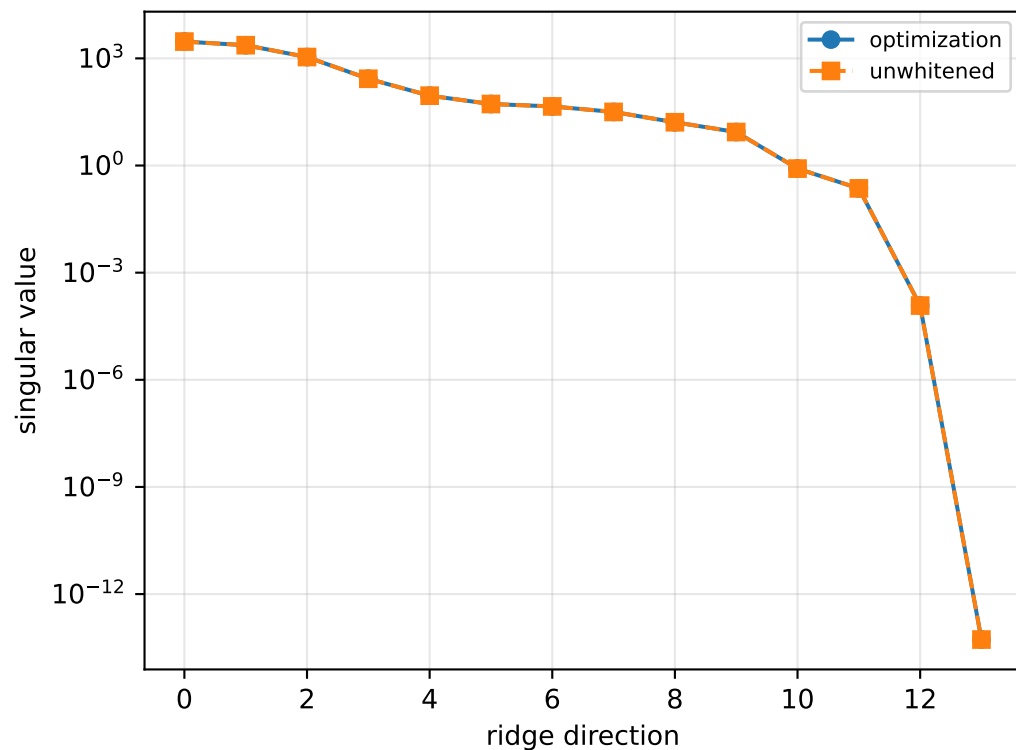
[-1.64915634e-02 9.52257991e-06 1.71547962e-01]]

scale-free f/m: [0.33490906 0.43126309 0.37659816 0.31223847]

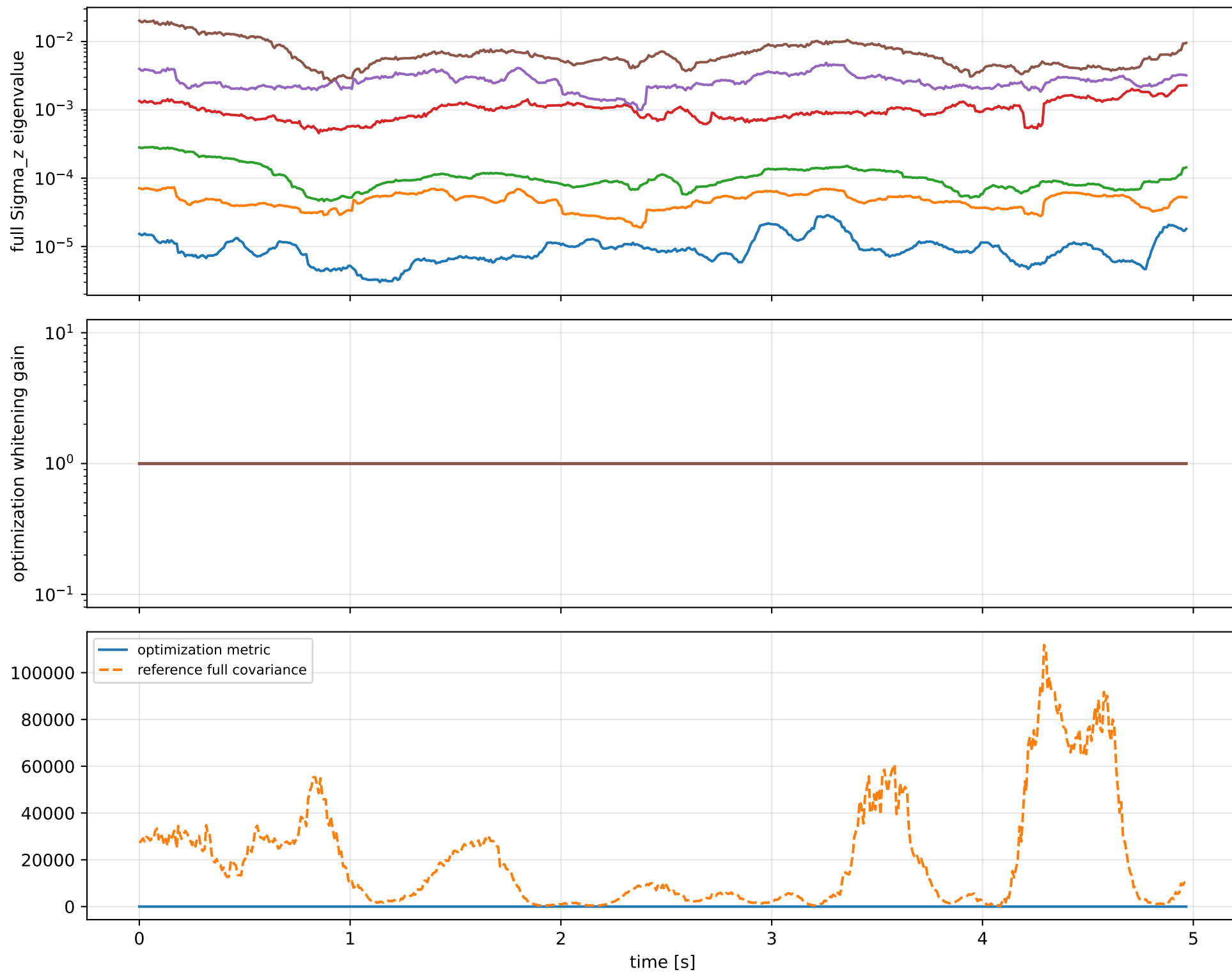
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

kkt_scale_positive: ridge spectrum (rank 13, nullity 1, ||Jv||=1.25e-13)



kkt_scale_positive: optimization vs reference covariance



kkt_scale_positive: wrench / closure (force max $1.29\text{e-}14$, torque max $7.63\text{e-}16$)

